1. Competitive inhibition
2. Noncompetitive inhibition
3. Uncompetitive inhibition.

9.3.1.1. Competitive inhibition (or) substrate analogue inhibition :

It is a type of reversible inhibition in which the inhibitor resembles the substrate and binds to the active site of the enzyme.

Since the competitive inhibitor resembles the substrate, it is otherwise known as substrate analogue.

In competitive inhibition, the enzyme can bind substrate (ES complex) or inhibitor (EI)

The inhibitor thus competes with the substrate to combine with the enzyme.

A competitive inhibitor decreases the rate of catalysis by reducing the proportion of enzyme molecules bound to a substrate.

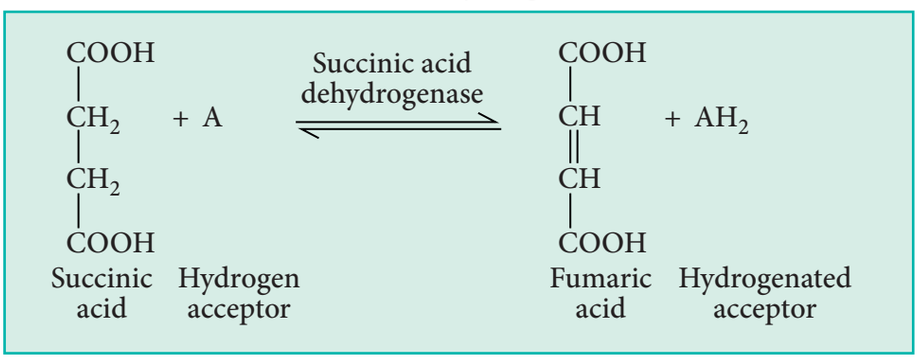
The degree of inhibition depends upon the relative concentrations of both substrate and inhibitor.

Competitive inhibition can be overcome by a sufficiently high concentration of substrate.

In competitive inhibition K_m value increases, whereas V_{max} remains unchanged.

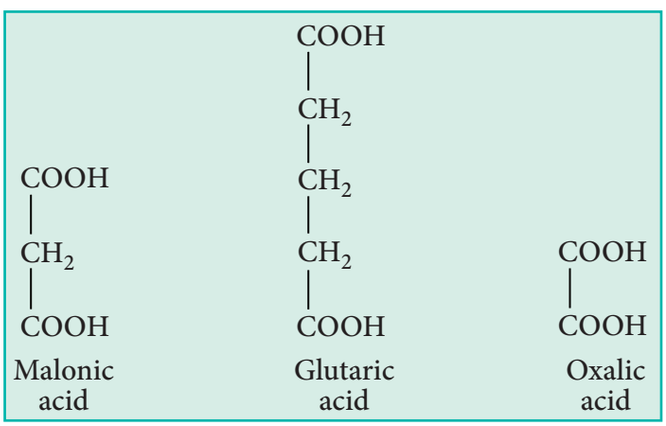
Example :

The action of malonate on succinate dehydrogenase (SDH)



Oxalic, malonic and glutaric acids are competitive inhibitors of SDH. Among them, malonic acid is the most potent competitive inhibitor.

The pharmacological action of many drugs may be explained by the principle of competitive inhibition. For example, sulphonamides are very commonly used antibacterial drugs.



XII U9 Enzyme Kinetics.indd 217 4/8/2020 3:27:14 PM